

Legacy vs corrected LEC climatology — technical report

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The rerun is complete: all 3,820 cyclones are validated.

1. What is compared

The rerun (`scripts/lec_climatology_rerun`) recomputes the semi-Lagrangian Lorenz Energy Cycle with `LorenzCycleToolKit 2.0.0`, which corrected Ca, the fifth Ck subterm, B Φ Z, B Φ E, vertical-level alignment, time tendencies and NaN handling. Everything else is held fixed: the same cyclones, the same frozen lifecycle windows, the same 3-hourly time steps, the same 15° × 15° moving box and the same 32-level (10–1000 hPa) control volume. Both sides are phase means over identical windows, so every difference reported here is attributable to the equation and numerical correction alone.

The legacy side is `data/energy_cache.parquet`, the article input. Re-aggregating the archived Zenodo per-cyclone results for 10 random cyclones reproduces that cache to a maximum relative difference of 2e-14, confirming the two sides are built the same way.

Paired sample: **3,820 cyclones**, 15,829 cyclone-phase rows, 24 terms.

2. What changed and what did not

16 of the 24 terms are numerically unchanged (relative change below 1%, Spearman ≈ 1 , essentially no sign changes): Az, Ae, Kz, Ke, Cz, Ce, Gz, Ge, BAz, BAe, BKz, BKe, $\partial Az/\partial t$, $\partial Ae/\partial t$, $\partial Kz/\partial t$ and $\partial Ke/\partial t$. This includes all four energy reservoirs, both generation terms, the four lateral boundary transports and all four budget tendencies — the corrections did not touch them.

The 8 terms that did change are exactly those the toolkit correction targets, plus the residuals that inherit them (see `figures/paired_control/violin_conversion.png` and `violin_boundary.png`; the energy, generation and budget figures show the two halves of each violin overlying each other exactly):

Term	Group	Median legacy	Median corrected	Median Δ	Relative change	Spearman	Sign changes
Ca	conversion	0.50	1.54	1.03	2.10×	0.97	5.1%
B Φ E	boundary	34.12	-1.11	-36.38	1.03×	-0.42	68.1%
B Φ Z	boundary	54.26	6.85	-43.86	0.87×	0.48	28.2%
RGe	residual	1.31	0.17	-1.02	0.57×	0.75	21.4%
Ck	conversion	-1.91	-0.03	1.81	0.53×	0.90	20.7%
RKz	residual	-0.99	0.20	1.03	0.42×	0.85	17.5%
RKe	residual	-5.29	-3.37	1.81	0.31×	0.98	7.5%
RKz	residual	8.18	6.04	-1.81	0.09×	1.00	3.0%

Conversion terms — legacy vs corrected LEC (3,820 cyclones)

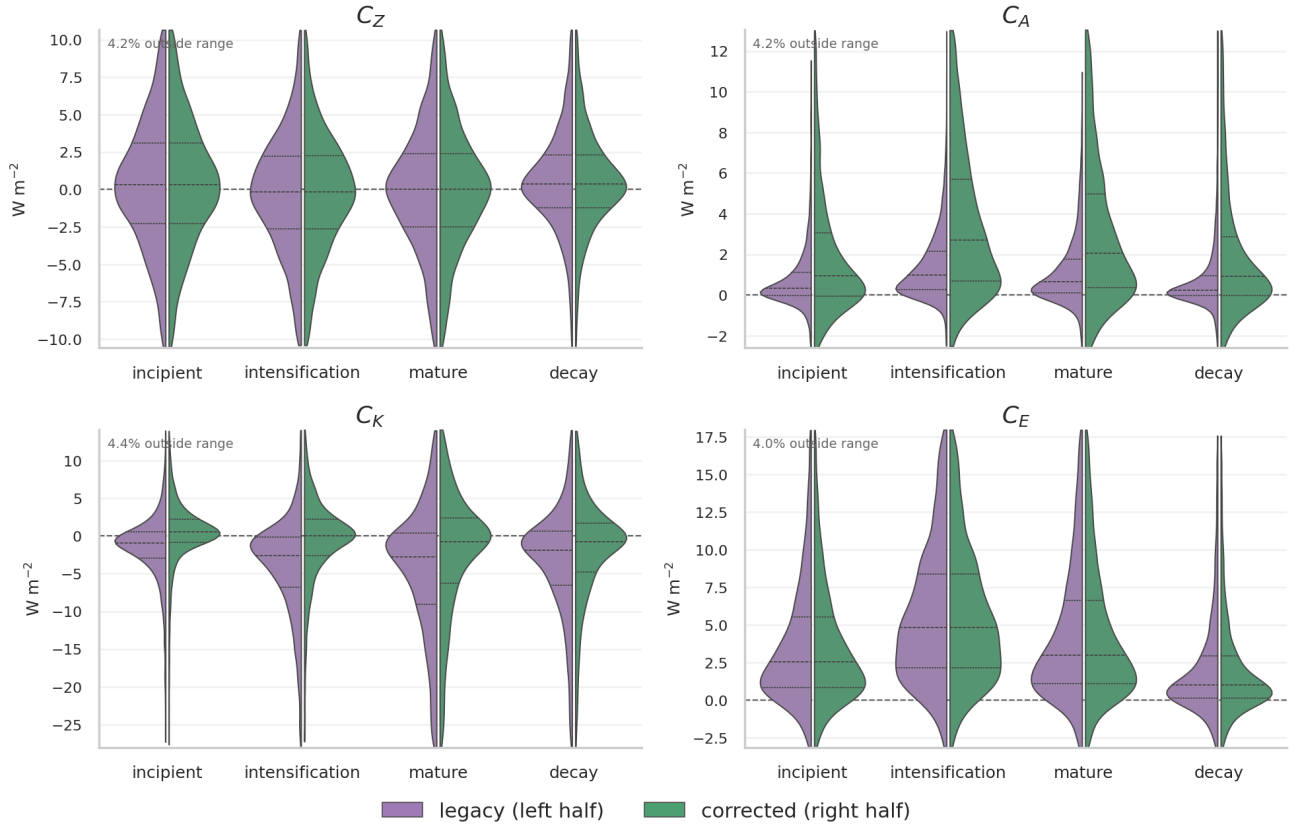


Figure 1. Conversion terms, legacy (left half) versus corrected (right half) of each violin, by life-cycle phase. C_A shifts upward in every phase and C_K crosses zero.

Boundary (transport) terms — legacy vs corrected LEC (3,820 cyclones)

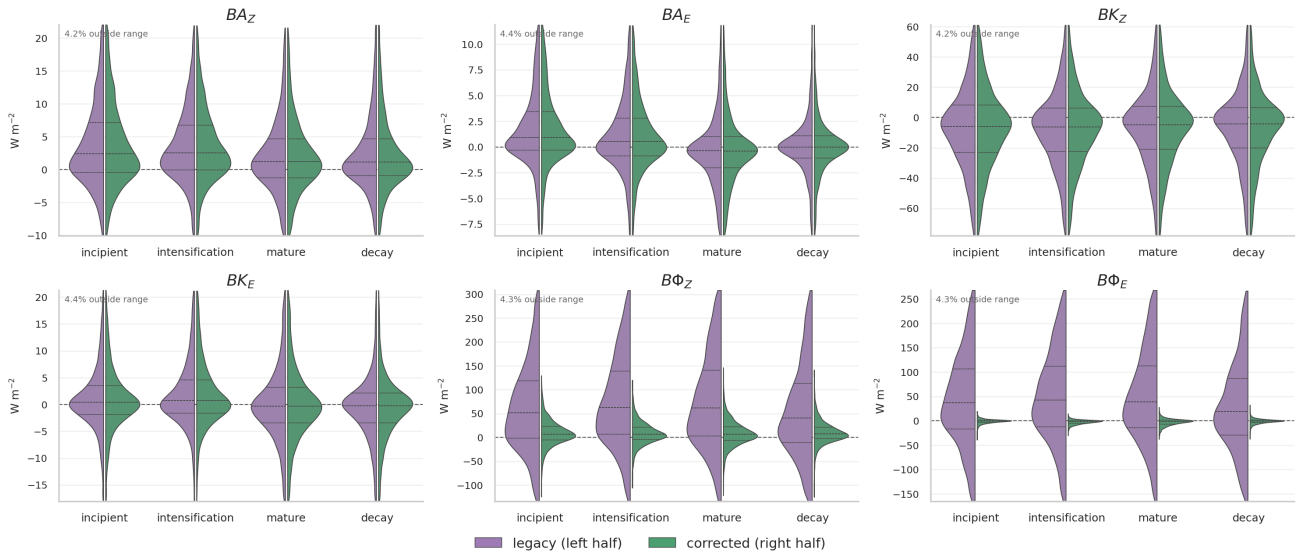


Figure 2. Boundary transport terms. The four lateral fluxes are unchanged; the two pressure-work terms $B\Phi_Z$ and $B\Phi_E$ collapse.

Residual terms — legacy vs corrected LEC (3,820 cyclones)

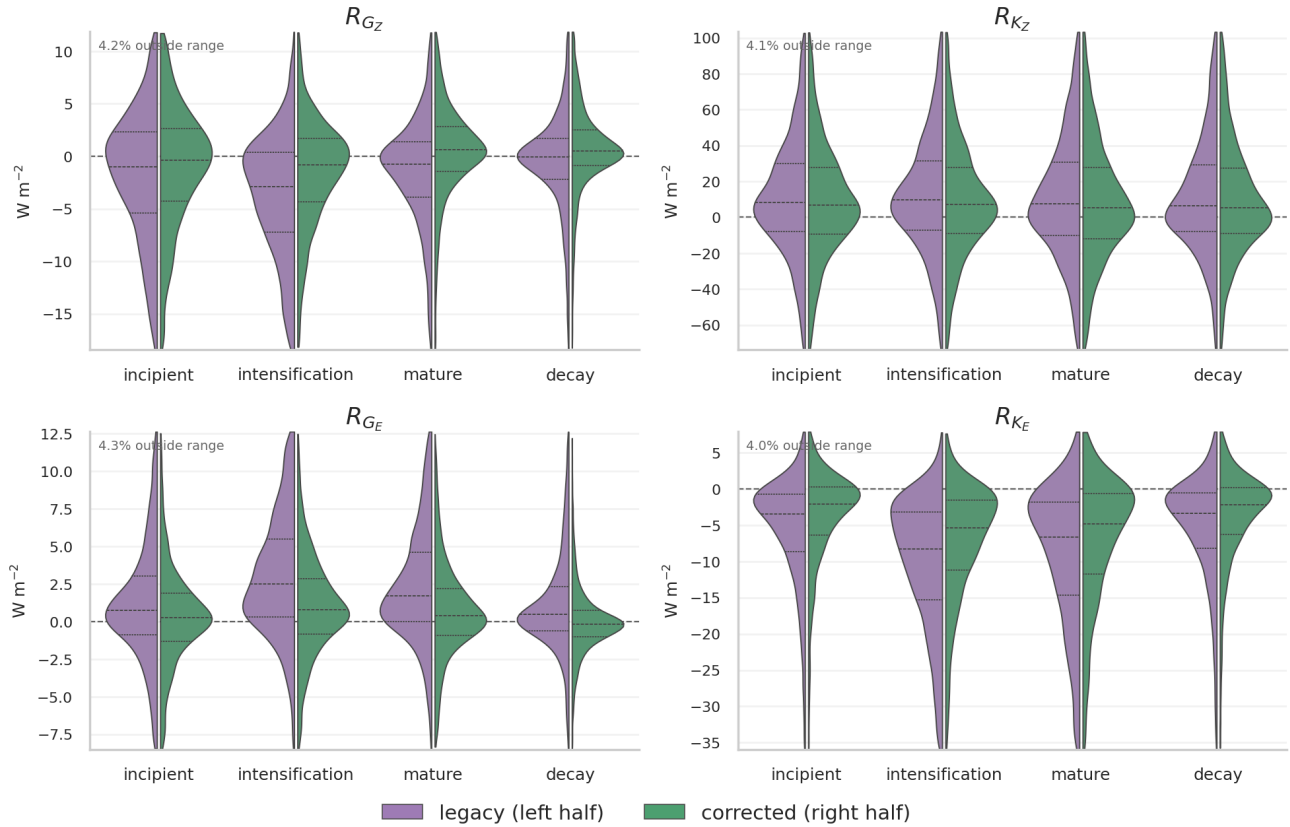


Figure 3. Residual terms, which inherit the C_A and C_K corrections through the budget closure.

The energy, generation and budget-tendency violins are not reproduced here: the two halves of every violin coincide exactly. They are in `figures/paired_control/` if a reader wants to confirm it.

Relative change is the median $|\Delta|$ divided by the median $|\text{legacy}|$ value: $1.00\times$ means the typical change is as large as the term itself. *Spearman* is the rank correlation between the two versions across cyclone-phases — the quantity that matters for the downstream PCA and k-means, which depend on ordering.

Three features stand out:

- Ca **roughly triples** (median $0.50 \rightarrow 1.54 \text{ W m}^{-2}$) but keeps its ranking (Spearman 0.97) and its sign in 95% of cases. Baroclinic conversion is systematically stronger, not reordered.
- Ck **shifts by about 1.81 W m^{-2}** , an almost uniform positive offset that carries the median from -1.91 to -0.03 W m^{-2} . The shift is largest in the intensification phase (2.96 W m^{-2}).
- **The pressure-work boundary terms collapse.** B Φ Z and B Φ E lose most of their amplitude, and B Φ E is now *anti*-correlated with its legacy counterpart (Spearman -0.42), so the legacy term cannot be rescaled into agreement — its structure was wrong, not just its amplitude.

The residuals R_{GZ} , R_{KZ} , R_{GE} and R_{KE} change by the same amounts as Ca and Ck with opposite signs, as expected from the budget closure — they are a consistency check rather than an independent finding. The corrected toolkit also outputs two terms with no legacy counterpart (C_{overturning}, M); they are not comparable and are excluded.

3. Sign changes

Sign is what carries physical meaning in the LEC, so a change of sign matters more than a change of magnitude. Per-sample sign-change rates are in `figures/paired_control/signflip_heatmap.png`; the

terms at risk are Ca, Ck, BΦZ, BΦE, RGz, RGe and RKe. The worst case is BΦE, which changes sign in 68% of cyclone-phases.

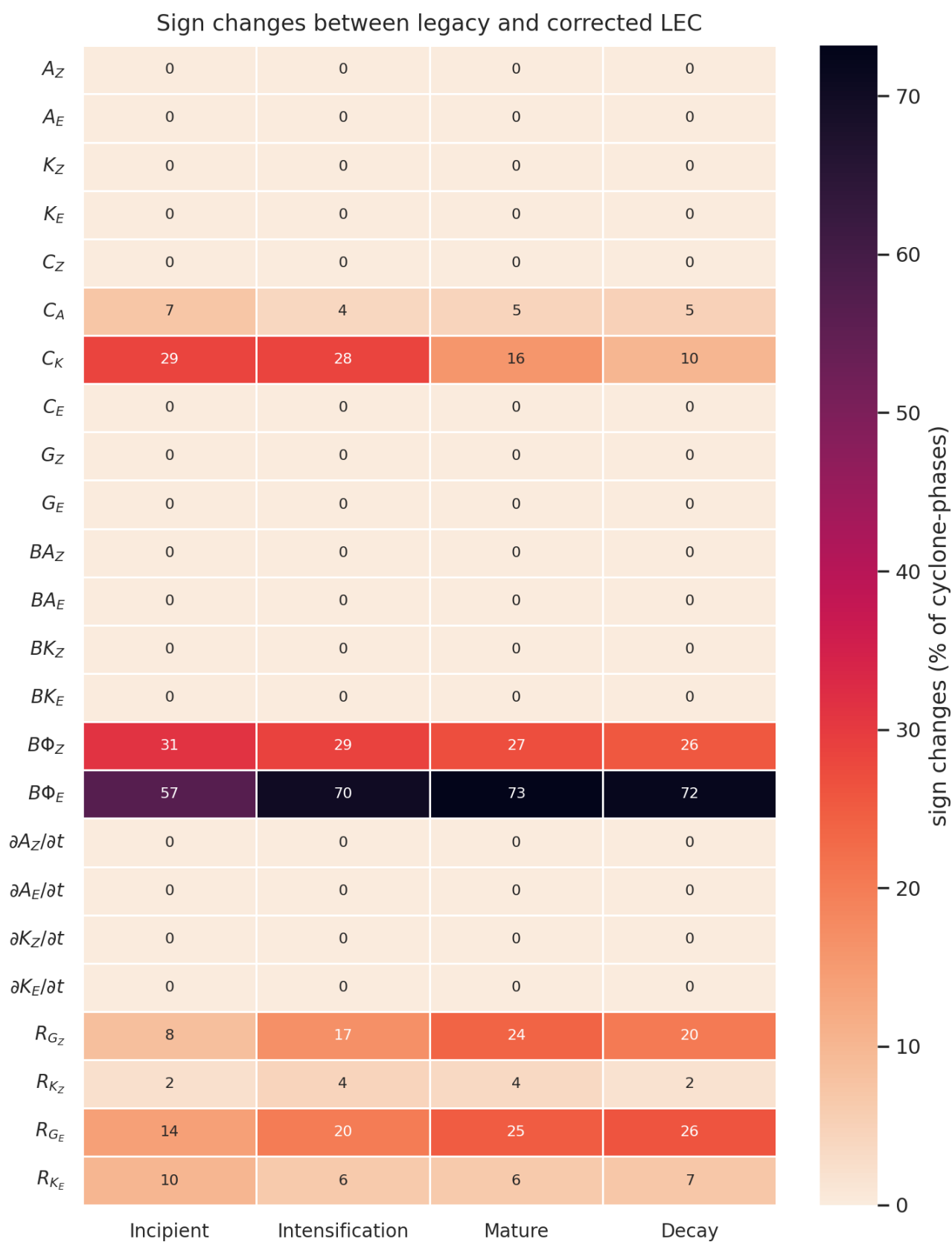


Figure 4. Percentage of cyclone-phases whose term changed sign between the two versions.

The terms that change the sign of their *pooled climatological median* are **BΦE and RGz**. Although the pooled Ck median remains slightly negative, its phase median changes sign during incipience

and intensification. Ck is the consequential clustering term because it enters the conversion LPS; BAE and RGz enter neither, appearing only in the all-terms effect-size figure (Fig. S2), which would need regenerating but carries no headline claim.

4. Consequences for the interpretation

The clustering input is only partly affected. The energy patterns are built from seven terms (Ca, Ck, BAe, BKe, Ae, Ke and Ge) across four phases. 5 of the seven (BAe, BKe, Ae, Ke and Ge) are unchanged; only Ca and Ck move. The EP separation should therefore survive in outline, but the two conversion terms are precisely the ones that define the conversion Lorenz Phase Space and the EP1 signature, so the clustering must be rerun before any EP statement is reasserted.

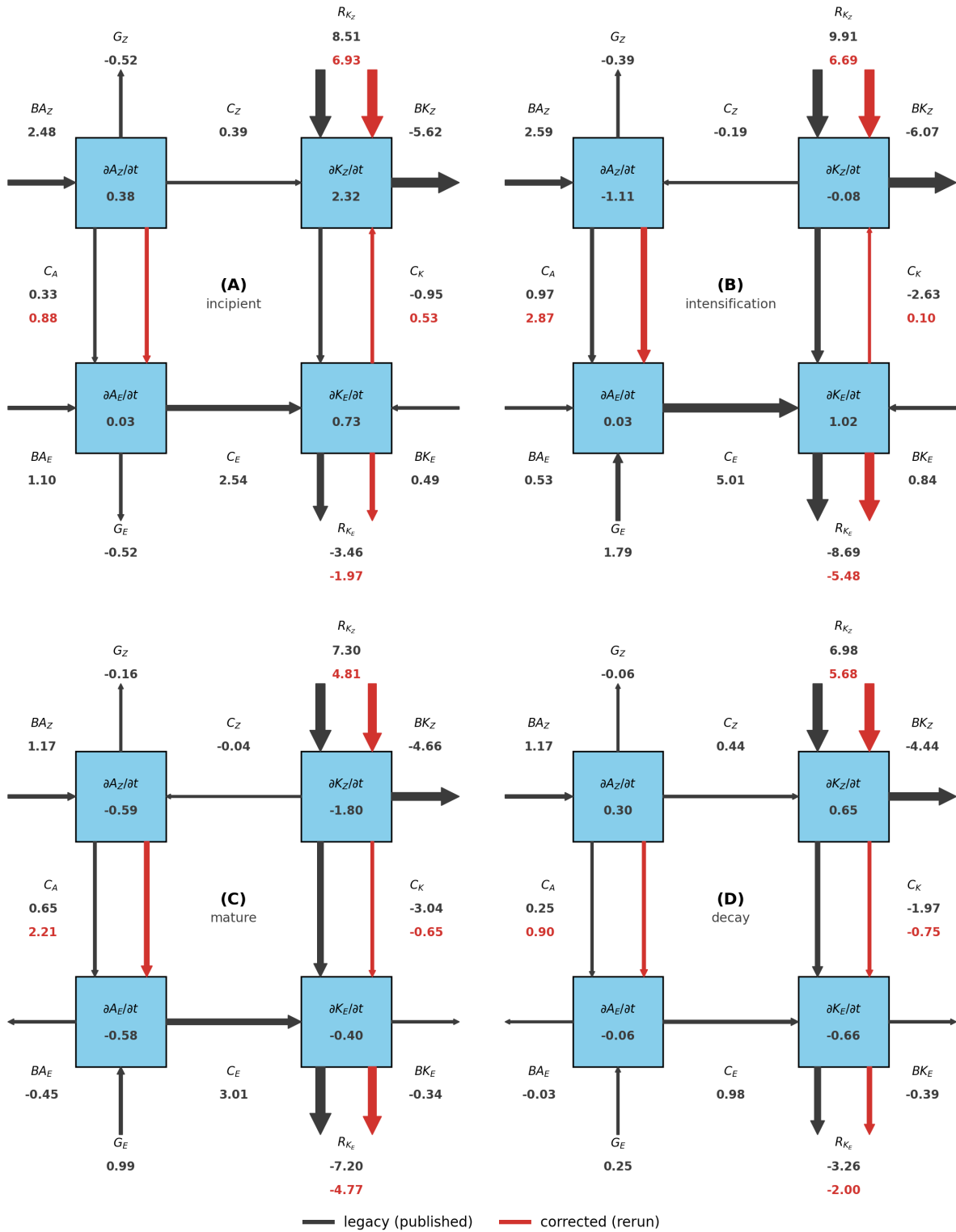
The barotropic result is the one that moves.

figures/paired_control/lec_diagram_before_after.png shows this on the four-box diagram: the Ck arrow reverses between the two versions during the incipient and intensification phases, and shortens during maturity and decay. With the convention of de Souza et al. (2025) — $Ca > 0$ feeds eddy APE, $Ck < 0$ feeds eddy KE:

Phase	Ck < 0 (legacy → corrected)	Ca > 0 & Ck < 0	Ck > Ca with Ck < 0	Median Ck
incipient	67% → 39%	53% → 30%	57% → 19%	-0.95 → 0.53
intensification	77% → 48%	69% → 43%	70% → 19%	-2.63 → 0.10
mature	71% → 55%	62% → 50%	66% → 37%	-3.04 → -0.65
decay	69% → 58%	58% → 51%	65% → 42%	-1.97 → -0.75
all phases	71% → 50%	61% → 43%	65% → 29%	-1.91 → -0.03

Pooled over the life cycle, the share of cyclone-phases with barotropic conversion feeding the eddy falls from 71% to 50%, occupancy of the doubly eddy-feeding quadrant falls from 61% to 43%, and the share of cases where barotropic conversion exceeds baroclinic conversion falls from 65% to 29%. During intensification the last figure drops from 70% to 19%.

Lorenz Energy Cycle before and after the toolkit correction
phase medians (W m^{-2}); doubled arrows mark terms that changed



Arrows point in the direction the energy flows; thickness scales with [median]. Unchanged terms carry a single arrow.

Changed terms with no place in the classical diagram (medians over all phases, W m^{-2}): $\text{B}\Phi\text{Z } 54.26 \rightarrow 6.85$ · $\text{B}\Phi\text{E } 34.12 \rightarrow -1.11$ · $\text{R}\text{Gz } -0.99 \rightarrow 0.20$ · $\text{R}\text{G}\text{e } 1.31 \rightarrow 0.17$

Figure 5. Four-box Lorenz Energy Cycle by life-cycle phase: (A) incipient, (B) intensification, (C) mature, (D) decay. Dark is legacy, red is corrected.

This weakens the article's headline claim that barotropic conversions are ubiquitous during cyclone evolution and can exceed baroclinic conversions. In the corrected climatology, barotropic conversion is close to neutral in the median, it is a minority contributor during intensification, and it feeds the eddy in roughly half of the cases rather than seven in ten. Two qualitative statements survive and

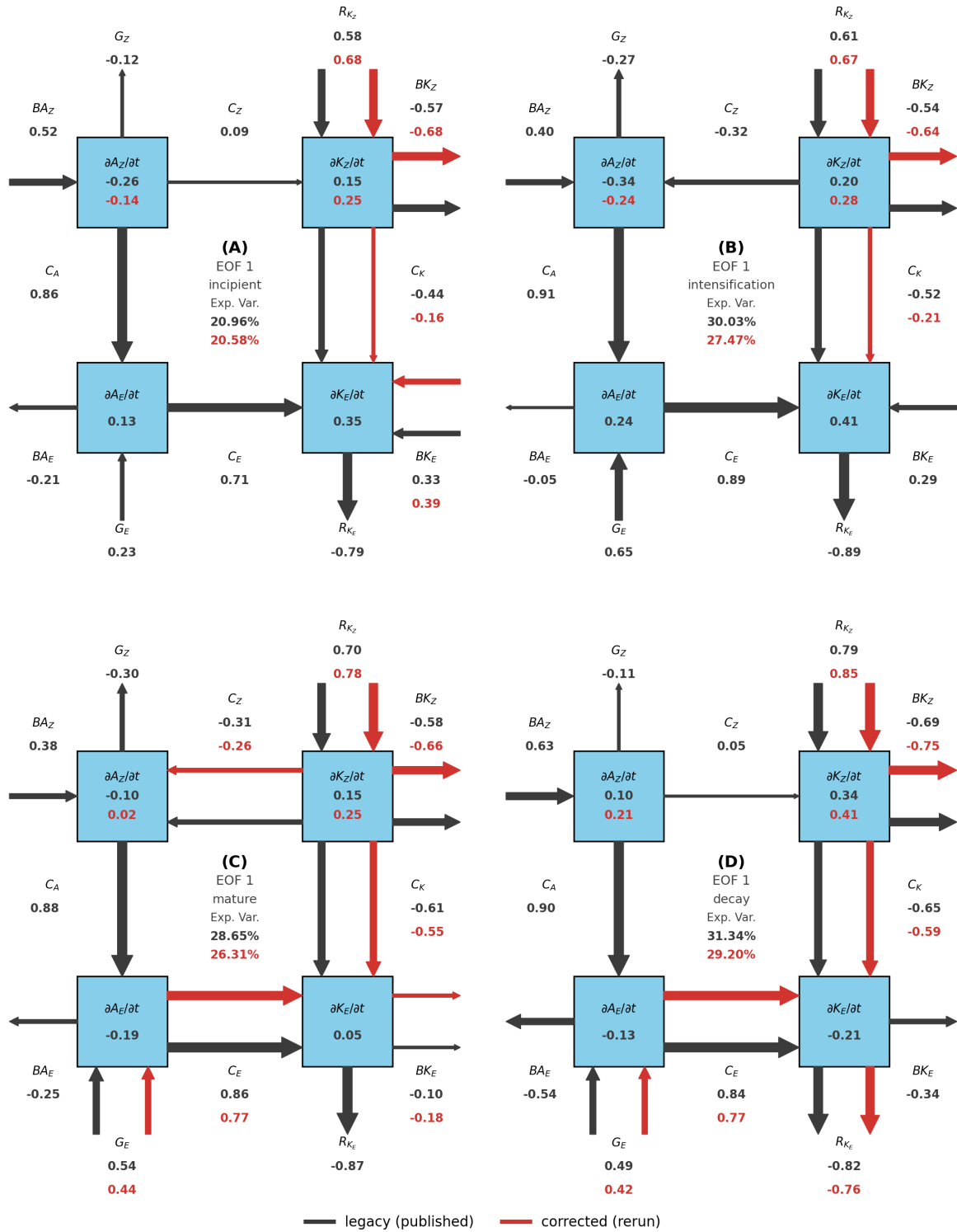
one does not:

- *Survives*: baroclinic conversion feeds the eddy in the large majority of cases, and is now unambiguously the dominant eddy source (Ca roughly tripled while Ck moved toward zero).
- *Survives, restated*: barotropic conversion remains present and eddy-feeding in a substantial minority of cyclone-phases, so it is still a real pathway — it is no longer the typical one.
- *Does not survive as stated*: the ordering of the phases. The legacy Ck was most strongly eddy-feeding during intensification and maturity; the corrected Ck is weakest during the incipient phase and becomes progressively more eddy-feeding toward decay. Any statement tying the barotropic peak to maturity needs to be rechecked against the corrected LPS.

The leading EOF survives in shape but is reweighted.

figures/paired_control/eof1_diagram_before_after.png redraws the thesis EOF figure with both versions on the same axes. EOF 1 still explains a comparable share of the variance (21-31% before, 21-29% after) and the two patterns correlate at 0.77-0.86 across phases, so the mode is recognisably the same. 7 of the 24 loadings move by less than 0.05, including BAz, $\partial Ae/\partial t$, Gz, Az and Ca. The largest shifts are B Φ E (1.43), RGe (0.67), RGz (0.61), i.e. the terms the correction hit directly; among the terms the classical diagram actually shows, the largest are Ck (0.31), $\partial Az/\partial t$ (0.12), BKz (0.11). The pattern is therefore stable where the underlying terms did not change and moves where they did — which is what the downstream PCA and k-means will inherit, and another reason to rerun them rather than assume the EPs carry over. Beyond the leading mode the rank itself is not preserved — legacy EOF 2 in the incipient phase matches corrected mode 3; legacy EOF 3 in the incipient phase matches corrected mode 2; legacy EOF 2 in the intensification phase matches corrected mode 4; legacy EOF 3 in the intensification phase matches corrected mode 2; legacy EOF 4 in the intensification phase matches corrected mode 3; legacy EOF 2 in the mature phase matches corrected mode 3; legacy EOF 3 in the mature phase matches corrected mode 2; legacy EOF 3 in the decay phase matches corrected mode 4; legacy EOF 4 in the decay phase matches corrected mode 3 — because EOF 2 and EOF 3 explain similar variance and the correction is enough to reorder them. The figures pair modes by pattern correlation rather than by rank; comparing them by rank would be misleading.

EOF 1 of the Lorenz Energy Cycle before and after the toolkit correction loadings (correlation with the PC); doubled arrows mark $|\Delta| \geq 0.05$



Arrows point in the direction implied by the sign of the loading; thickness scales with $|\text{loading}|$. Both modes are recomputed on the paired sample and sign-aligned.
B Φ Z, B Φ E, RGZ, RGE take part in the EOF but have no place in the classical diagram; their loadings are in results/lec_rerun_comparison/eof_loadings.csv (largest change: B Φ E +0.61 $\rightarrow$ -0.82 during mature).

Figure 6. EOF 1 loadings on the LEC diagram, by phase: (A) incipient, (B) intensification, (C) mature, (D) decay. Dark is legacy, red is corrected.

The same diagram for EOF 2 to EOF 4 (Figures 6 to 8 of the article) is in eof2_diagram_before_after.png, eof3_... and eof4_....

The Bø collapse does not affect the published figures (the article uses BAe and BKe for the import LPS, both unchanged), but it does affect the budget closure and any future use of the pressure-work terms.

5. Caveats

- The population is complete.
- Statistics are medians and rank correlations throughout, because LEC phase means are heavy tailed. The violin figures are trimmed for display only; every number here uses the full sample.
- Secondary lifecycle periods (decay 2, etc.) are matched to their own legacy counterpart and folded into the parent phase, as in the corrected cache builder. Rows named `residual` are excluded.
- This report compares diagnostics, not conclusions. Nothing here substitutes for rerunning the PCA, the k-means and the EP figures on the corrected cache.

6. Files

Artifact	Path
This report (PDF, figures embedded)	<code>docs/paired_control/lec_rerun_paired_control_report.pdf</code>
Paired table	<code>results/paired_control/paired_terms.parquet</code>
Term summary	<code>results/paired_control/term_change_summary.csv</code>
Per-phase summary	<code>results/paired_control/term_change_by_phase.csv</code>
Conversion regime	<code>results/paired_control/conversion_regime.csv</code>
Coverage / provenance	<code>results/paired_control/coverage.json</code>
Split-violin figures	<code>figures/paired_control/violin_{energy,conversion,generation,boundary,budget,residual}.png</code>
Sign-change heatmap	<code>figures/paired_control/signflip_heatmap.png</code>
Before/after LEC diagram	<code>figures/paired_control/lec_diagram_before_after.png</code>
Before/after EOF diagrams	<code>figures/paired_control/eof{1,2,3,4}_diagram_before_after.png</code>
EOF loadings and variance	<code>results/paired_control/eof_loadings.csv, eof_variance.csv</code>